

Dr. Ankit Barik

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Experience

Dept. of Earth & Planetary Sciences, Johns Hopkins University **Baltimore, USA**
Assistant Research Scientist *Nov 2022 – Present*

- Research on fluid interiors of planets and icy moons, planetary magnetic fields and stellar interiors.
- Research software engineering (NASA Space Technology Research Institute IMQCAM) for next-generation physics-based workflows for model-based qualification and certification of additive manufacturing of spacecraft parts.

Dept. of Earth & Planetary Sciences, Johns Hopkins University **Baltimore, USA**
Postdoctoral researcher *Nov 2017 – Nov 2022*

Max Planck Institute for Solar System Research **Göttingen, Germany**
Postdoctoral researcher *May 2017 – Oct 2017*

Education

International Max Planck Research School for Solar System Science **Göttingen, Germany**
PhD, Magna cum laude *2013 – 2017*

- Thesis title: Inertial modes, turbulence and magnetic effects on a differentially rotating spherical shell
- Thesis supervisors: Dr. Johannes Wicht, Prof. Dr. Ulrich R. Christensen, Prof. Dr. Andreas Tilgner
- Defence date: 08 May, 2017

Indian Institute of Technology, Kharagpur **Kharagpur, India**
Bachelor's + Master's *2008 – 2013*

- *Major:* Exploration Geophysics, *Minor:* Physics
- Thesis title: Effect of gravity environment on dynamo action in rotating spherical shells
- Thesis supervisors: Dr. Johannes Wicht, Prof. W.K. Mohanty

Summer/Winter Schools.....

- 12th International School/Symposium for Space Simulations (ISSS-12), Prague, Czech Republic, July 2 - 6, 2015
- 'Turbulence, magnetic fields and self organization in laboratory and astrophysical plasmas', Les Houches, France, March 23 - April 03, 2015

Mission involvement

Part of science team of the InSight (Interior Exploration using Seismic Investigations, Geodesy and Heat Transport) Mars mission.

Code development

- **MagIC** : 3D pseudo-spectral magnetohydrodynamics (MHD) code to study planetary and stellar interiors. Community code used in over 100 publications. (<https://github.com/magic-sph/magic>)
- **Kore** : 3D spectral MHD eigenvalue code. (<https://github.com/repepo/kore>)
- **Kaiju** : 3D finite volume MHD code to study magnetospheres (<https://kaiju-docs.readthedocs.io/>)
- **planetMagFields** : Teaching/research tool to visualize magnetic fields of planets in our solar system. (<https://ankitbarik.github.io/planetMagFields>).
- **inermodz** : Python package to compute and plot analytical inertial eigenmodes of a sphere (<https://github.com/AnkitBarik/inermodz>).
- **Additive manufacturing** : finite-element code for additively manufactured alloys (research software engineering for NASA IMQCAM), data/workflow pipelines through use of Dagster and Globus compute.

Grants/Awards

Visiting positions.....

2024 Invited professorship at École Centrale Méditerranée/IRPHE, Marseille, France. Total funding received : \$10,000

Research Grants.....

2024 Collaborator, NSF Astronomy and Astrophysics Research Grants (AAG)
2020 (Co-wrote) NASA grant proposal for the Cassini Data Analysis Program : \$488,710

Computing time.....

2026 Co-PI, NSF ACCESS computing time (ID: PHY260120): 23.5k node-hours
2026 Co-PI, NSF ACCESS computing time (ID: PHY260012): 23.5k node-hours
2025 PI, NSF ACCESS computing time (ID: EES250080): 23.5k node-hours
2024 PI, NSF ACCESS computing time (ID: EES240070): 1.5 million cpu-hours
Jul 2015 - Jul 2020 Computational time at the North-German Supercomputing Alliance : \$600,000

Teaching.....

2023 Cloos Memorial Lecturer, Department of Earth & Planetary Sciences, Johns Hopkins University : \$7,535.

Other.....

2021 Postdoctoral science teaching fellowship for course "Stellar & Planetary Waves"
2013 Among top 1% selected for a program for fully sponsored PhD in computational sciences by Shell. (declined)
2013 Best Master's Thesis, Department of Geology & Geophysics, IIT Kharagpur
2012 Ranked 2nd in India in Schlumberger's coding competition for a seismic inversion plug-in for their software 'Petrel'

Proposals submitted, not funded.....

2025 PI, NASA Solar System Science (S3)
2025 Co-I, NASA Interdisciplinary Consortia for Astrobiology Research (ICAR)
2024 Co-I, NASA Cassini Data Analysis Program (CDAP)
2022 Subaward, NSF Astronomy and Astrophysics Research Grants (AAG)

Teaching

March 2021 Certificate of completion - Johns Hopkins "Teaching Academy"
○ Attending course "Preparation for university teaching"
○ Attending pedagogy seminars/workshops
○ More than six hours of teaching
2024 Sep - Nov Invited professor at École Centrale Méditerranée / IRPHE, Marseille, France

Graduate courses.....

2026 Spring Guest lecturer, "Planetary Interiors", Johns Hopkins University
2024 Nov Guest lecturer "Environmental Fluid Mechanics", Centrale Méditerranée, Marseille, France
2024 Oct Guest lecturer "Numerical methods: spectral methods", Centrale Méditerranée, Marseille, France
2024 Fall Guest lecturer "Magnetic Fields in Planetary Systems", Purdue University
2023 Fall Cloos Memorial Lecturer "Earth and Planetary Fluids", Johns Hopkins University
2023 Spring Guest lecturer, "Planetary Interiors", Johns Hopkins University

2021 Spring	Guest lecturer , "Planetary Interiors", Johns Hopkins University
2021 Spring	Guest lecturer , "Special topics in dynamo theory", Johns Hopkins University
2019 Fall	Guest lecturer , "Earth and Planetary Fluids I", Johns Hopkins University
2019 Spring	Guest lecturer , "Planetary Interiors", Johns Hopkins University
2014 Fall	Teaching assistant , "Solar System Science: The Central Star", University of Göttingen

Undergraduate courses.....

2014 Spring	Teaching assistant , "Computational Physics", University of Göttingen
2014 Spring	Teaching assistant , "Introduction to Astro-and Geophysics", University of Göttingen

Other.....

2015 Nov 4-6	Tutor , hands-on workshop on 'MagIC' code "Dynamos in a Nutshell"
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Mentoring and supervision

Graduate students.....

- Hachem Dhouib, PhD student at CEA Saclay, for Kavli Summer Program in Astrophysics 2021, June 7th to July 16th, 2021
 - Project: Angular momentum transport by gravito-inertial waves in intermediate-mass stars
- PhD students in the research group: Chi Yan (graduated), Regupathi Angappan (graduated)

Undergraduate students.....

Fall 2018	Nina Amezcua	Exoplanet magnetic fields
Fall 2018	Mackenzie Mills	Ancient martian dynamo
Summer 2020	Brian Song	(co-advising) Magnetic data from Iridium Satellites
Summer 2021	Nick Lu	(co-advising) Magnetospheric simulations of the Earth
Summer 2021	Vishnu Srinivasan	(co-advising) Spherical harmonic transforms, use of MagIC simulation code

Professional services

Proposal review

- NSF, 2026
- NASA (2026, 2020, 2019)
- Swiss National Supercomputing Centre (CSCS) (2026, 2025)
- Czech Science Foundation, 2024
- ETH Zurich, (Sep & May 2022)

PhD thesis review and defense committee

Albert Elias López, Institute of Space Sciences (ICE-CSIC)/Universitat Autònoma de Barcelona, Barcelona, Spain

Journal referee

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| ○ Nature | ○ Astronomy & Astrophysics |
| ○ Nature Geosciences | ○ Planetary Science Journal |
| ○ Comptes Rendus Physique | ○ Geophysical Journal International |
| ○ European Journal of Mechanics - B/Fluids | ○ Earth, Planets and Space |
| ○ Geophysical & Astrophysical Fluid Dynamics | ○ Geophysical Research Letters |
| ○ Earth and Space Science | ○ International Journal on Geomathematics |
| ○ The Astrophysical Journal | ○ Journal of Geophysical Research: Planets |
| ○ Journal of Open Source Software | ○ Physics of Fluids |
| ○ Space Science Reviews | |

Member

- Executive committee, web / social media manager for Geomagnetism, Paleomagnetism and Electromagnetism (GPE) Section of AGU
- American Geophysical Union (AGU)
- American Physical Society (APS)
- AAS Division for Planetary Sciences (DPS)

Conference organisation.....

2024, Dec 9-13	Convener, session “P023 - Oscillations in Internal Fluid Layers of Planets, Moons, and Stars” at AGU Fall Meeting 2024
2024, Dec 9-13	Convener, session “DI003 - Core-Mantle Interactions: The Dynamic Duo Shaping Our Planet” at AGU Fall Meeting 2024
2023, Dec 12	Convener, session “P23G - Oscillations in Internal Fluid Layers of Planets, Moons, and Stars” at AGU Fall Meeting 2023
2016, Nov 30-Dec 2	17 th MHD Days, 88 participants
2015, Nov 22-24	14 th General meeting of PhDnet, 99 participants
2015, Nov 4-6	MagIC code workshop “Dynamoes in a Nutshell”, 35 participants

Outreach

2025, Sep 16	“Magnetic fields of the planets around us”, Lecturers Without Borders (LeWiBo)
2023, Apr 4	AGU “Ask a Scientist” for Earth Day 2023, Washington DC
2020 - present	Social media manager for DIYnamics (Twitter: @DIYnamicsTeam) - an outreach effort from UCLA for studying/demonstrating geophysical fluid dynamics at home/class
2019, Sep/Oct	Outreach video “The Magnetic Fields of the Solar System” (https://www.youtube.com/watch?v=7S_VqFJep_0) - 44k views
2019, Oct 8	Talk “Everything wrong with The Core” - a talk on the movie
2019, Jul 24	Talk “Planetary magnetic fields: where do they all come from?” at the 2019 QuarkNet workshop
2018	Volunteer at the National Air and Space Museum, Washington DC

Publications

Articles.....

- [1] L. Ojha, **A. Barik**, and J. Buffo. Limited direct fluid exchange between the deep subsurface ocean and the shallow subsurface environment of Europa. *Nature Astronomy*, July 2026.
- [2] J. R. Fuentes, **A. Barik**, and J. Fuller. Excitation of Inertial Modes in 3D Simulations of Rotating Convection in Planets and Stars. *The Astrophysical Journal*, 998(1):131, February 2026.
- [3] C. Yan, **A. Barik**, S. Stanley, A. Mittelholz, A.-C. Plesa, and C.-L. Johnson. Mars' hemispheric magnetic field from a full-sphere dynamo. *Geophysical Research Letters*, 52(3):e2024GL113926, 2025.
- [4] **A. Barik**, S. A. Triana, M. Hoff, and J. Wicht. Transition to turbulence in the wide-gap spherical Couette system. *Journal of Fluid Mechanics*, 1001:A1, December 2024.
- [5] **A. Barik** and R. Angappan. planetmagfields: A python package for analyzing and plotting planetary magnetic field data. *Journal of Open Source Software*, 9(97):6677, 2024.
- [6] F. Seuren, S. A. Triana, J. Rekier, **A. Barik**, and T. Van Hoolst. Effects of the Librationally Induced Flow in Mercury's Fluid Core with an Outer Stably Stratified Layer. *The Planetary Science Journal*, 4(9):161, September 2023.
- [7] C. Yan, **A. Barik**, S. Stanley, J. Leung, A. Mittelholz, C. L. Johnson, A.-C. Plesa, and A. Rivoldini. An ancient martian dynamo driven by hemispheric heating: effect of thermal boundary conditions. *Planetary Science Journal*, 2023.
- [8] T. Gastine, **A. Barik**, rraynaud, t schwaiger, B. Putigny, thtassin, J. Wicht, L. Duarte, and B. Dintrans. magic-sph/magic: release MagIC 6.2, December 2022.
- [9] **A. Barik**, S. A. Triana, M. Calkins, Stanley S., and J. Aurnou. Onset of convection in rotating spherical shells: Variations with radius ratio. *Earth and Space Science*, 2022.
- [10] K. M. Moore, **A. Barik**, S. Stanley, D. J. Stevenson, N. Nettelmann, R. Helled, T. Guillot, B. Militzer, and S. Bolton. Jupiter's dynamo magnetic field: The role of stable stratification and a dilute core. *Journal of Geophysical Research: Planets*, 2022.
- [11] S. A. Triana, G. Guerrero, **A. Barik**, and J. Rekier. Identification of inertial modes in the solar convection zone. *The Astrophysical Journal Letters*, July 2022.
- [12] B. J. Anderson, R. Angappan, **A. Barik**, S. K. Vines, S. Stanley, P. N. Bernasconi, H. Korth, and R. J. Barnes. Iridium Communications Satellite Constellation Data for Study of Earth's Magnetic Field. *Geochemistry, Geophysics, Geosystems*, August 2021, **Highlighted by Nature magazine** (<https://www.nature.com/articles/d41586-021-01860-9>).
- [13] V. Perera, C. Mead, K. J. van der Hoeven Kraft, S. Stanley, R. Angappan, S. MacKenzie, **A. Barik**, and S. Buxner. Considering intergroup emotions to improve diversity and inclusion in the geosciences. *Journal of Geoscience Education*, July 2021.
- [14] **A. Barik**, S. A. Triana, M. Hoff, and J. Wicht. Triadic resonances in the wide-gap spherical couette system. *Journal of Fluid Mechanics*, 2018.

Book Chapters.....

- [15] M. Le Bars, **A. Barik**, F. Burmann, D. P. Lathrop, J. Noir, N. Schaeffer, and S. A. Triana. Fluid Dynamics Experiments for Planetary Interiors. *Surveys in Geophysics*, December 2021.

Submitted/Under review.....

- [16] J Rekier, S. A. Triana, Abdulah D. **Barik**, **A.**, and W. Kang. Nutation damping from core-mantle boundary topography.

Soon to be submitted.....

- [17] **A. Barik**, S. Stanley, B. Tian, S. Tikoo, and B. Weiss. An ancient lunar dynamo driven by mantle precession and convection.
- [18] **A. Barik**, J. M. Aurnou, and T. Gastine. Convection at the poles: a study across planetary bodies.
- [19] R. Angappan, **A. Barik**, B. J. Anderson, S. K. Vines, and Stanley S. Fast global wave detection in geomagnetic jerk occurrences with commercial satellites.
- [20] S. A. Triana, J. Rekier, F. Gerick, **A. Barik**, and V. Dehant. Torsional Alfvén modes in the earth's core: A numerical model.

Selected talks and Posters

Invited talks at conferences/workshops.....

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| 2024, Aug 26-30 | Effect of the Mantle on Planetary Core-driven Dynamos, <i>Magnetism, Atmospheres, and Life workshop</i> , University of Illinois Urbana-Champaign |
| 2022, Dec 12-16 | Comparison of Jupiter's and Saturn's magnetic fields and implications for their interiors, <i>AGU Fall Meeting 2022</i> , Chicago |
| 2022, Jul 14 | Effect of libration on a stable layer: an application to Mercury, <i>17th SEDI symposium</i> , ETH Zurich |
| 2021, Jul 27 | planetMagFields : A python package for planetary magnetic fields, <i>OpenPlanetary Virtual Lunches</i> , Virtual |
| 2021, Jul 08 | Inertial modes from differential rotation: the spherical Couette system, <i>Kavli Summer Program in Astrophysics 2021</i> |
| 2020, Sep | Dynamos driven by convection and precession, <i>17th Symposium of Study of the Earth's Deep Interior (SEDI)</i> , Virtual |
| 2020, Sep 1-4 | Triadic resonances in the spherical Couette system, <i>ISSI workshop on Deep Earth</i> , Bern, Switzerland |
| 2017, Feb 27-28 | Inertial and magneto-Coriolis modes in the spherical Couette flow, <i>3rd ANR IMAGINE Meeting</i> , L'Institut de Recherche en Astrophysique et Planétologie (IRAP), Toulouse, France |
| 2014, Dec 13 | Spherical Couette flow simulations, <i>Workshop on Geomagnetic Prediction</i> , hosted by the CIDER project, UC Berkley, Berkley, California, USA |

Invited seminars.....

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| 2025, March 4 | Invited seminar at Southwest Research Institute (SwRI), Boulder, Colorado, USA |
| 2025, Feb 25 | Center for Environmental and Applied Fluid Mechanics (CEAFM) seminar, Johns Hopkins University, USA |
| 2024, Nov 12 | Seminar at Institut de Recherche sur les Phénomènes Hors Equilibre, Marseille |
| 2024, Nov 4/5 | Hosted by Henri-Claude Nataf at ISTERre, Grenoble |
| 2024, Oct 21/22 | Hosted by Hagay Amit at University of Nantes |
| 2024, Sep | <i>Geophysical/Astrophysical Fluid Dynamics Seminar</i> , University of Colorado Boulder |
| 2023, Oct 11 | <i>Planetary Lunch Series</i> , Massachusetts Institute of Technology |
| 2022, Oct 27 | <i>Fluids & MHD seminar</i> , University of Leeds |
| 2022, May 27 | <i>IGPP Seminar</i> , UC Santa Cruz |
| 2021, Nov 11 | <i>EPM Group Seminar</i> , ETH Zurich |
| 2018, Feb 15 | <i>Applied Dynamics Seminar Series</i> , University of Maryland, College Park, USA |

Contributed talks.....

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| 2021, Dec 13-17 | Onset of convection in rotating spherical shells, <i>AGU Fall Meeting 2021</i> |
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2020, Dec 1-17 The ancient lunar dynamo, *AGU Fall Meeting 2020*, Virtual
2019, Dec 9-13 Inertial Wave Generation from Boundary Layer Turbulence, *AGU Fall Meeting 2019*, San Francisco, USA
2019, May 20-22 A Lunar dynamo driven by mantle precession and convection, *Core of the Moon workshop*, Marseille, France
2017, Jun 25-Jul 1 Triadic resonances in the spherical Couette flow, *2nd Conference on Natural Dynamos*, Valtice, Czech Republic
2017, Jun 25-Jul 1 Spherical Couette dynamos, *2nd Conference on Natural Dynamos*, Valtice, Czech Republic
2015, Jun 22-24 Flow instabilities in the Spherical Couette System, *19th International Couette-Taylor Workshop*, Brandenburg University of Technology, Cottbus, Germany

Contributed posters.....

2025, Dec 15-19 Rotating convection with topography: Implications for Core-Mantle and ice-ocean boundaries, *AGU Fall Meeting 2025*, New Orleans, USA
2025, Sep 7-12 Topography at the ice-ocean boundary: effects on heat transport, *EPSC-DPS Joint Meeting 2025*, Helsinki, Finland
2025, Jan 27-31 Topography at the core-mantle boundary: effects on heat transport, *IPAM workshop on Rotating Turbulence: Interplay and Separability of Bulk and Boundary Dynamics*, UCLA, Los Angeles, USA
2024, Dec 12 planetMagFields: A Python package for analyzing and visualizing planetary magnetic field data, *AGU Fall Meeting 2024*, Washington DC, USA
2024, Jun 12-13 Interaction of Convection and Tides in Icy Moons, *OPAG Meeting*, Ithaca, NY, USA
2023, Nov 19-21 Kore : A spectral anelastic MHD eigenvalue code for rotating fluids in spherical geometries, *76th Annual Meeting of the Division of Fluid Dynamics*, Washington DC, USA
2018, Dec 10-14 A Lunar Dynamo Driven by Mantle Precession and Convection, *AGU Fall Meeting 2018*, Washington DC, USA
2018, Jul 8-13 Turbulence in spherical Couette flow and the effect of density stratification, *Study of the Earth's Deep Interior (SEDI) 2018*, Edmonton, Canada
2016, Dec 12-16 Identification and onset of inertial modes in the wide-gap spherical Couette system, *AGU Fall Meeting 2016*, San Francisco, USA